Benchmark Report for the `bispectrum` Library

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1 Overview

This document presents computational benchmarks for the `bispectrum` library, a PyTorch implementation of the group bispectrum for invariant signal representations. The library provides six distinct modules, each implementing the bispectrum for a different group acting on a different space:

- `CnonCn` — cyclic group C_n acting on itself;
- `TorusOnTorus` — product of cyclic groups $C_{n_1} \times \cdots \times C_{n_d}$ (torus $\mathbb{T}^d$) acting on itself;
- `DnonDn` — dihedral group D_n acting on itself;
- `SO2onD2` — continuous rotation group SO(2) acting on the 2-disk D^2 (via truncated Bessel basis);
- `SO3onS2` — rotation group SO(3) acting on the 2-sphere S^2 (via spherical harmonics);
- `OctaonOcta` — octahedral group O (order 24) acting on the octahedron.

Each module is a `torch.nn.Module` exposing a `forward(f)` method that computes the bispectrum of an input signal f, and (where available) an `invert(beta)` method that recovers a signal from its bispectral coefficients up to the group action.

A key design axis is *selective* versus *full* bispectrum: the full bispectrum enumerates all $O(|G|^2)$ pairwise products of Fourier coefficients, while the selective bispectrum retains only the $O(|G|)$ coefficients needed for injectivity, following the theoretical framework of Kakarala (2012).

All benchmarks were run on a single NVIDIA A100 80 GB PCIe GPU with PyTorch 2.10.

2 Coefficient Scaling

Discussion. Figure 1 confirms that the selective bispectrum achieves the theoretically predicted $O(|G|)$ coefficient count across all module families, providing a quadratic reduction over the full $O(|G|^2)$ variant. For the cyclic group at $|G| = 1024$, this translates from 524 800 full coefficients down to 1 024 selective coefficients—a 512× reduction. The D_n selective counts lie slightly above the $O(|G|)$ reference because D_n has more irreducible representations than C_n at the same group order (due to the semidirect product structure), but the scaling remains linear. The continuous-group modules (SO(2)/D^2 and SO(3)/S^2) exhibit comparable linear scaling when measured against the square of their truncation bandwidth, confirming that the selective strategy extends beyond finite groups.

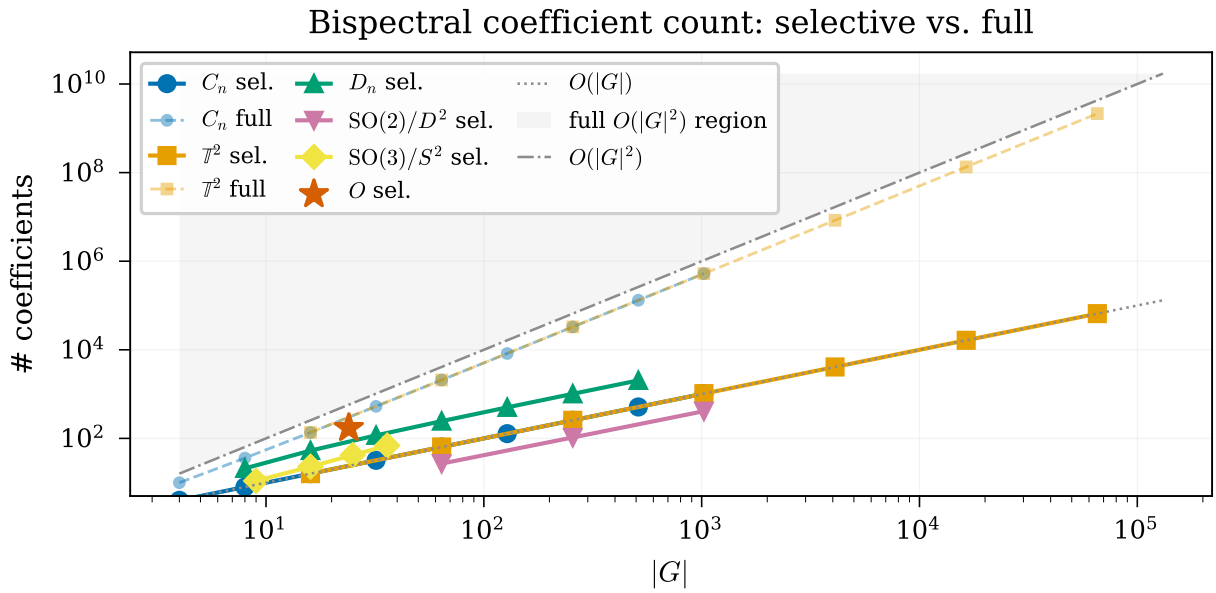

Figure 1: Number of bispectral coefficients as a function of group order $|G|$ for all six modules. Solid lines with filled markers: selective bispectrum; dashed lines: full bispectrum (shown for C_n and $\mathbb{T}^2$ only, the two modules that support it). The gray $O(|G|)$ and $O(|G|^2)$ reference slopes are shown, with the shaded region indicating where the full bispectrum lies. All selective curves track the $O(|G|)$ slope. The continuous-group modules (SO(2)/D^2 and SO(3)/S^2) also follow the linear trend when plotted against their effective bandwidth parameter. The octahedral group O ($|G| = 24$, star) yields 172 selective coefficients.

3 Forward-Pass Timing

Discussion. Figure 2 demonstrates that the selective forward pass is *effectively free* on modern GPUs: the wall-clock time is dominated by kernel-launch overhead rather than arithmetic, staying flat at $\sim$0.1–0.2 ms even as the group order increases by two orders of magnitude. The full bispectrum, in contrast, exhibits the expected super-linear growth and rapidly becomes impractical for production use. For the torus at $|G| = 1{,}024$ ($n = 32$), the full forward pass is already 5$\times$ slower than selective; extrapolating the quadratic trend to $n = 256$ would exceed 100 ms per sample.

4 GPU Batch Scaling and Inversion

Discussion. Figure 3(a) shows that all selective-bispectrum modules scale linearly with batch size on the GPU, which is the ideal behavior for integration into batched training pipelines. The throughput spans roughly three orders of magnitude across modules, reflecting the underlying signal dimension: C_{128} and $\mathbb{T}^2$ ($|G| = 1024$) process each sample in microseconds, while O is much slower per sample due to the structure of the octahedral bispectrum computation. The key takeaway is that *none of the modules exhibit throughput degradation at large batch sizes*, confirming that the implementation is memory-bandwidth-bound rather than compute-bound at these scales.

Figure 3(b) reveals that inversion cost varies significantly across modules. For C_n and D_n, inversion is essentially a linear-time operation (inverse FFT plus phase retrieval), scaling as $O(|G|)$. The SO(2)/D^2 module shows similar linear scaling despite operating over a continuous group (the

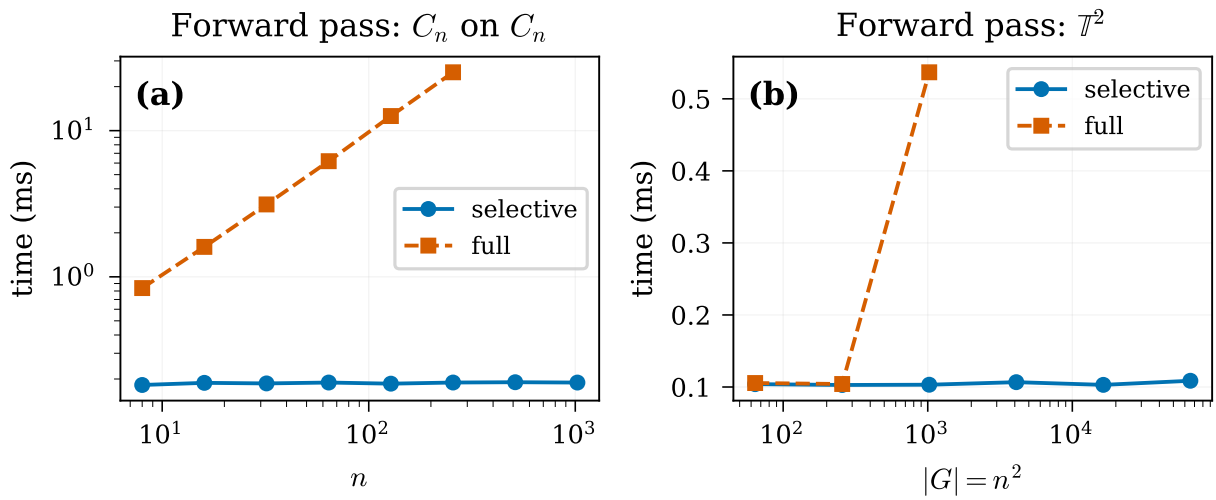

Figure 2: Forward-pass wall-clock time (batch size 16, NVIDIA A100 GPU) for the selective versus full bispectrum. **(a)** C_n on C_n: the selective forward pass stays flat at $\sim$0.19 ms regardless of n, while the full bispectrum grows super-linearly to 25 ms at $n = 256$. **(b)** $\mathbb{T}^2 = C_n \times C_n$: the selective pass is nearly constant ($\sim$0.1 ms) up to $|G| = 65{,}536$, while the full bispectrum diverges beyond $|G| \approx 1{,}000$.

truncated Bessel basis provides a finite-dimensional approximation). The torus module's inversion is more expensive, growing as $O(|G| \log |G|)$ because it involves multi-dimensional inverse FFTs. The octahedral group is an outlier: its inversion requires $\sim$1 s for a group of order 24, reflecting the combinatorial complexity of inverting over the 5 non-trivial irreducible representations of O and the associated least-squares optimization.

5 Summary Table

Table 1 summarizes key data points for each module at a representative group size. The selective coefficient count is always $O(|G|)$: for C_{128}, it equals $|G| = 128$; for D_{32} ($|G| = 64$), the 245 coefficients reflect the richer representation theory of the dihedral group. $\mathrm{SO}(2)/D^2$ at $L = 32$ (bandwidth 1024) yields 411 selective coefficients and requires $\sim$12 ms for a forward pass, dominated by the Bessel-function evaluations in the radial basis. $\mathrm{SO}(3)/S^2$ at $\ell_{\max} = 5$ produces 69 coefficients with a comparable forward time; its inversion is not yet implemented.

The "full" columns are only applicable to the abelian groups (C_n and $\mathbb{T}^2$), where the full bispectrum is the classical definition. For non-abelian groups (D_n, O) and continuous groups (SO(2), SO(3)), only the selective bispectrum is implemented, as the full enumeration would be either ill-defined or computationally intractable.

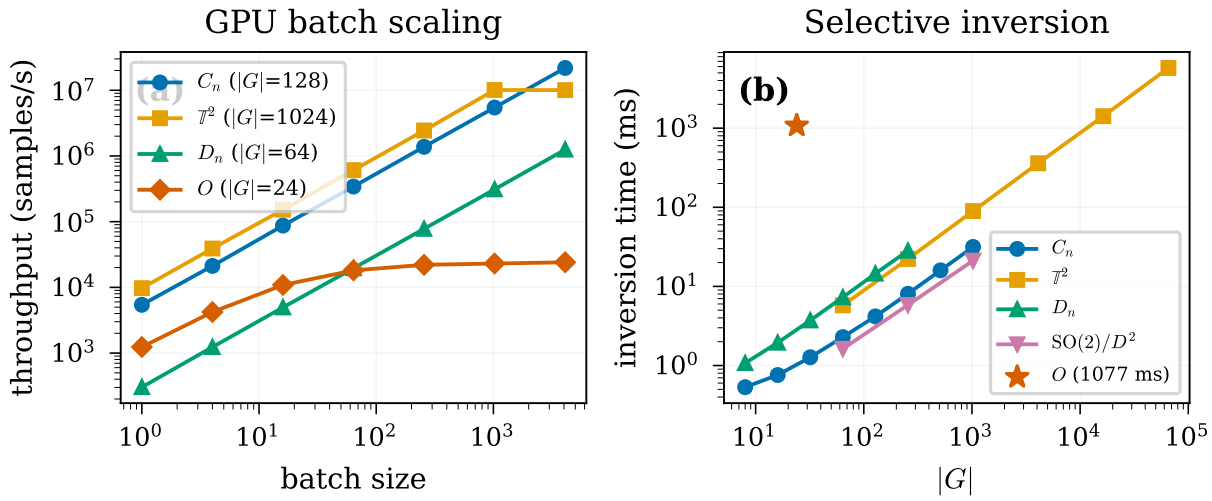

Figure 3: **(a)** GPU throughput (samples/s) as a function of batch size for the selective forward pass at representative group sizes. All modules exhibit near-linear scaling, confirming the selective bispectrum is well suited to batched deep-learning workloads. C_n and $\mathbb{T}^2$ achieve over 10^7 samples/s. The octahedral group plateaus at $\sim$24 k samples/s due to its fixed small signal dimension ($|G| = 24$) and CUDA launch overhead. **(b)** Inversion wall-clock time for all modules supporting `invert()` (batch size 4). C_n, D_n, and SO(2)/D^2 scale roughly linearly. $\mathbb{T}^2$ grows as $O(|G| \log |G|)$ from multi-dimensional inverse FFTs. The octahedral group (star) requires $\sim$1 s despite $|G| = 24$ due to the least-squares optimization over all irreducible representations of O. `SO3onS2` is absent because its inversion is not yet implemented.

Table 1: Benchmark summary for the `bispectrum` library. Coefficient counts compare the selective ($O(|G|)$) and full ($O(|G|^2)$) bispectra. Timings are median wall-clock on a single NVIDIA A100 GPU (batch = 16 for forward, 4 for inversion). "–" indicates the mode is not implemented.

Module	G	$\|G\|$	Coefs		Fwd (ms)		Inv (ms)
			Sel.	Full	Sel.	Full	Sel.
`CnonCn`	C_{128}	128	128	8256	0.19	12.50	4.18
`TorusOnTorus`	$C_{32}\times C_{32}$	1024	1024	524800	0.10	0.54	88.79
`DnonDn`	D_{32}	64	245	–	3.24	–	7.42
`SO2onD2`	SO(2)	1024	411	–	12.21	–	22.29
`SO3onS2`	SO(3)	36	69	–	11.07	–	–
`OctaonOcta`	O	24	172	–	1.48	–	4599.82